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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Loan Eligibility Verification and EMI Payment Tracking System

Submitted by

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DECLARATION

I hereby declare that the project entitled “Loan Eligibility Verification and EMI Payment Tracking System**”** submitted in partial fulfillment of the requirements for the course java programming is a **record of my original work** carried out under the guidance of Dr.S.Boopalan , Technical Head, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

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(BELSHIBA.J)

CERTIFICATE

This is to certify that the project entitled **"Loan Eligibility Verification and EMI Payment Tracking System"** is a **bona fide work** carried out by **BELSHIBA J**, I-B.Sc.CS -A[2025-2028], Department of Computer Science, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course JAVA PROGRAMMING LANGUAGE.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore**.**

Date:10.04.2026

Submitted for the Viva-Voce Examination held on 10.04.2026.

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ABSTRACT

The Loan Eligibility Verification and EMI Payment Tracking System is an advanced web-based application developed to automate and simplify the loan processing system. In traditional methods, loan approval and EMI tracking involve manual verification, which is time-consuming and prone to errors. This system aims to overcome such challenges by providing a fast, accurate, and user-friendly solution.

The system verifies loan eligibility based on multiple criteria such as user income, employment status, credit score, age, and existing financial obligations. By using predefined algorithms and conditions, it ensures fair and consistent decision-making. Users can easily input their details, and the system instantly determines whether they are eligible for a loan and the maximum loan amount they can avail.

Once the loan is approved, the system automatically generates an EMI schedule based on the loan amount, interest rate, and tenure. It provides a clear breakdown of monthly payments, including principal and interest components. Users can view their EMI details anytime through a dashboard.

The EMI tracking module helps users monitor their payment history, due dates, and outstanding balance. It also sends automated reminders through notifications or emails to ensure timely payments. In case of delayed payments, the system calculates penalties and updates the records accordingly.

Additionally, the system maintains a secure database to store user and transaction details, ensuring data integrity and privacy. It also provides administrative features for managing users, verifying applications, and generating reports.

Overall, the system enhances transparency, reduces manual effort, minimizes errors, and improves user experience for both financial institutions and borrowers. It serves as a reliable solution for streamlining loan processing and ensuring disciplined repayment management.

1. INTRODUCTION

The Loan Eligibility Verification and EMI Payment Tracking System is a modern solution developed to simplify the process of loan management and repayment tracking. In today's fast-paced financial environment, individuals frequently apply for loans for various purposes such as education, housing, or personal needs. However, traditional loan processing systems often involve manual verification and calculations, which can be time-consuming, inefficient, and prone to human errors.

This system is designed to automate the verification of loan eligibility by analyzing key financial parameters such as income, expenses, credit score, and repayment capacity. By using predefined rules and conditions, the system ensures accurate and quick decision-making regarding loan approval. This reduces the workload of financial institutions and enhances the experience for users.

In addition to eligibility verification, the system provides an effective way to calculate and track Equated Monthly Installments (EMIs). EMI calculation is a critical aspect of loan management, and the system ensures precise computation based on loan amount, interest rate, and tenure. Once the loan is approved, users can easily monitor their payment schedules, due dates, and outstanding balances.

The system also helps users maintain financial discipline by offering reminders and maintaining payment records. It serves as a centralized platform where all loan-related information is stored and managed efficiently. By integrating eligibility verification and EMI tracking into a single system, the project aims to provide a comprehensive and user-friendly solution.

Overall, this system demonstrates the effective use of technology in improving financial services, reducing errors, saving time, and enhancing transparency in loan management processes.

1.1 Objective

The main objective of the Loan Eligibility Verification and EMI Payment Tracking System is to automate the process of checking loan eligibility based on user financial details. It aims to reduce manual errors and improve the accuracy of loan approval decisions. The system is designed to calculate EMI efficiently using standard formulas based on loan amount, interest rate, and tenure.

Another objective is to help users track their EMI payments, including due dates, payment status, and outstanding balance. It also focuses on providing a user-friendly interface for easy interaction. The system ensures secure storage of user and loan data. Additionally, it aims to send reminders for timely payments and generate reports for better financial management.

1.2 Overview

The Loan Eligibility Verification and EMI Payment Tracking System is a software application designed to streamline loan-related processes for users and institutions. It provides a platform where users can input their financial details to check their eligibility for a loan quickly and accurately. The system uses predefined criteria such as income, expenses, and credit score to determine eligibility.

In addition to verification, the system calculates the EMI based on the loan amount, interest rate, and repayment period. It also allows users to track their EMI payments, including due dates and payment history. The application maintains a structured database to store all relevant information securely.

Overall, the system offers a simple, efficient, and reliable way to manage loan eligibility and EMI tracking in one place.

1.3 FEATURES

- Accurate EMI calculation using loan amount, interest rate, and tenure
- Easy tracking of EMI payments and due dates
- Payment history and outstanding balance management
- User-friendly interface for smooth interaction
- Secure storage of user and loan information
- Reminder notifications for upcoming EMI payments
- Quick generation of loan and payment reports

2. PROGRAM ANALYSIS

The Loan Eligibility Verification and EMI Payment Tracking System is developed to address the challenges faced in manual loan processing and repayment management. Traditional methods often involve lengthy procedures, paperwork, and a higher risk of calculation errors. This system analyzes user financial data such as income, expenses, and credit score to determine loan eligibility quickly and accurately.

It also evaluates the user's repayment capacity to ensure responsible lending decisions. The system integrates EMI calculation and tracking features, allowing users to monitor their payment schedules efficiently. By automating these processes, it reduces time consumption and improves overall accuracy.

Additionally, the program ensures proper data management by storing user and loan details in an organized manner. It enhances transparency and provides users with clear information about their loan status and repayment progress. Overall, the system improves efficiency, reliability, and user experience in loan management.

2.1 Problem Definition

In the existing system, loan eligibility verification and EMI tracking are often performed manually, which leads to delays and inaccuracies. Financial institutions and users face difficulties in calculating loan eligibility due to complex criteria involving income, expenses, and credit history. Manual EMI calculations can result in errors, affecting repayment planning and financial stability.

Users also lack a proper system to track their EMI payments, due dates, and outstanding balances, which may lead to missed payments and penalties. There is limited transparency in the loan approval process, making it harder for users to understand their eligibility status. Additionally, maintaining records manually is time-consuming and prone to data loss.

Therefore, there is a need for an automated system that can efficiently verify loan eligibility, accurately calculate EMIs, and provide a reliable way to track and manage loan repayments.

2.2 Project Overview

The process of verifying loan eligibility and managing EMI payments is often complex and inefficient in traditional systems. It typically involves manual calculations and document verification, which consume time and increase the chances of human error. Users may find it difficult to understand whether they qualify for a loan due to unclear criteria and lack of instant feedback.

Additionally, EMI calculation and tracking are not always handled systematically, leading to confusion about payment schedules and outstanding balances. Many users fail to keep proper records of their payments, which can result in missed deadlines and financial penalties. Financial institutions also face challenges in maintaining large volumes of data accurately.

Due to the absence of a centralized system, both users and institutions experience difficulties in managing loan-related information. Hence, there is a need for an automated and user-friendly system to simplify eligibility verification and EMI tracking processes.

2.3 Scopes and Limitations

Scope :

The Loan Eligibility Verification and EMI Payment Tracking System aims to automate and simplify the process of loan management. It helps users check their loan eligibility quickly and accurately based on financial data such as income, expenses, and credit score. The system calculates EMIs, tracks payment schedules, and maintains a history of transactions. It can generate reports and provide reminders for upcoming payments, improving financial discipline and transparency. The system is suitable for both individual users and small financial institutions.

Limitations:

The system depends on accurate input data from users for correct eligibility verification and EMI calculation. It may not cover complex banking rules or integrate with all financial institutions. The system is limited to predefined eligibility criteria and cannot make subjective decisions. Internet or software issues may affect accessibility, and security risks must be managed to protect sensitive financial information.

3. SYSTEM DESIGN

The **Loan Eligibility Verification and EMI Payment Tracking System** is designed to provide a structured and automated approach to loan management. The system design consists of multiple interconnected modules that handle eligibility verification, EMI calculation, and payment tracking.

3.1 Modules

The main components are as follows:

1. User Interface Module

- Provides a simple and interactive interface for users to input personal and financial details.
- Allows users to view loan eligibility status, EMI schedules, and payment history.

2. Loan Eligibility Verification Module

- Analyzes user data such as income, expenses, credit score, and loan amount requested.
- Compares the data against predefined eligibility criteria.
- Generates an eligibility result (approved or not approved).

3. EMI Calculation Module

- Calculates EMI using the standard formula based on loan amount, interest rate, and tenure.
- Provides detailed payment schedules showing principal and interest breakdown.

4. EMI Tracking and Payment Module

- Tracks all EMI payments and updates status (paid, due, or overdue).
- Maintains a complete record of all transactions.
- Sends reminders for upcoming EMI payments to avoid penalties.

5. Database Module

- Stores user details, loan information, EMI schedules, and payment history securely.
- Ensures data integrity and provides quick retrieval for reports and analysis.

6. Reporting Module

- Generates reports on loan eligibility, EMI status, and payment history.
- Helps users and institutions monitor loan management effectively.

System Flow:

1. User registers and logs into the system.
2. User enters financial details and requests loan eligibility check.
3. System verifies eligibility and provides result.
4. If approved, EMI is calculated and schedule generated.
5. System tracks payments, updates records, and sends reminders.

4. METHODOLOGY

The methodology for the Loan Eligibility Verification and EMI Payment Tracking System outlines the step-by-step approach used to analyze, design, develop, and implement the system. The methodology ensures systematic development and efficient functioning.

1. Problem Analysis

- Identify the challenges in manual loan verification and EMI tracking.
- Understand user requirements, including eligibility criteria, EMI calculation, and payment monitoring.
- Analyze the limitations of existing systems and define objectives for improvement.

2. System Design

- Create the system architecture and design the database structure.
- Define the modules: User Interface, Loan Eligibility Verification, EMI Calculation, EMI Tracking, and Reporting.
- Design flowcharts, algorithms, and user interaction processes for each module.

3. Program Development

- Write code for each module using appropriate programming languages and tools.
- Integrate all modules to create a cohesive system.
- Implement database connections and ensure secure data storage.

4. Testing and Validation

- Test individual modules for functionality and correctness.
- Validate the system with real or sample data to ensure accurate loan eligibility checks and EMI calculations.
- Identify and fix errors or inconsistencies in the system.

5. Result and Documentation

- Document system performance, outputs, and results from testing.
- Provide user manuals or guides for using the system effectively.

6. Algorithm

- Input user financial details (income, expenses, credit score).
- Verify eligibility based on predefined criteria.
- Calculate EMI if eligible.
- Store loan and payment information.
- Track EMI payments and update status.

7. Flowchart

Start

|

User Login

|

Enter Financial Details

|

Check Loan Eligibility

|

Eligible? ---- No ----> Display "Not Eligible" ----> End

|

Yes

|

Calculate EMI

|

Store Loan & EMI Data

|

Track EMI Payments & Update Status

|

End

5. SOURCE CODE

```
import java.util.Scanner;

 class   {

    String name;
    double income;
    double loanAmount;
    double interestRate = 8.0; // 8% interest
    int months;

    void getDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Customer Name: ");
        name = sc.nextLine();

        System.out.print("Enter Monthly Income: ");
        income = sc.nextDouble();

        System.out.print("Enter Loan Amount: ");
        loanAmount = sc.nextDouble();

        System.out.print("Enter Loan Duration (months): ");
        months = sc.nextInt();
    }
```

```
void checkEligibility() {
    if (income >= 20000) {
        System.out.println("Loan Approved for " + name);
    } else {
        System.out.println("Loan Not Approved (Income too low)");
    }
}

double calculateEMI() {
    double r = interestRate / (12 * 100);
    double emi = (loanAmount * r * Math.pow(1 + r, months)) /
                 (Math.pow(1 + r, months) - 1);
    return emi;
}

void showEMI() {
    double emi = calculateEMI();
    System.out.println("Monthly EMI: " + emi);
}

void trackPayment() {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter number of EMIs paid: ");
    int paid = sc.nextInt();

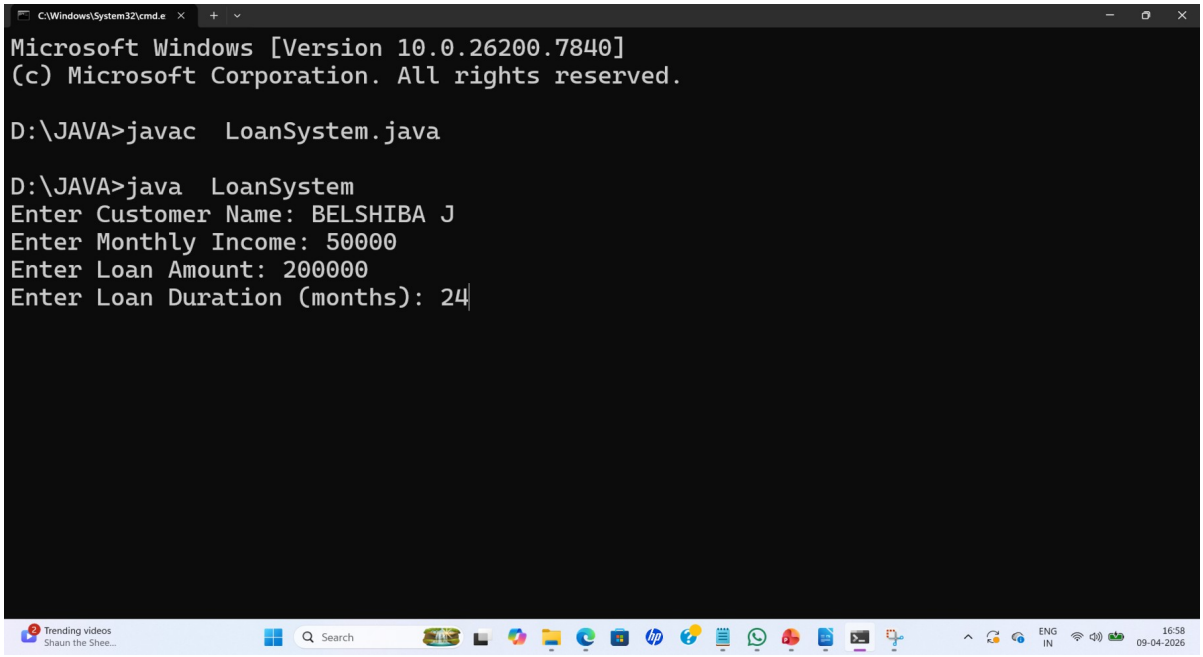
    int remaining = months - paid;
        System.out.println("Remaining EMI Months: " + remaining);
}
```

```
    public static void main(String[] args) {
        LoanSystem obj = new LoanSystem();

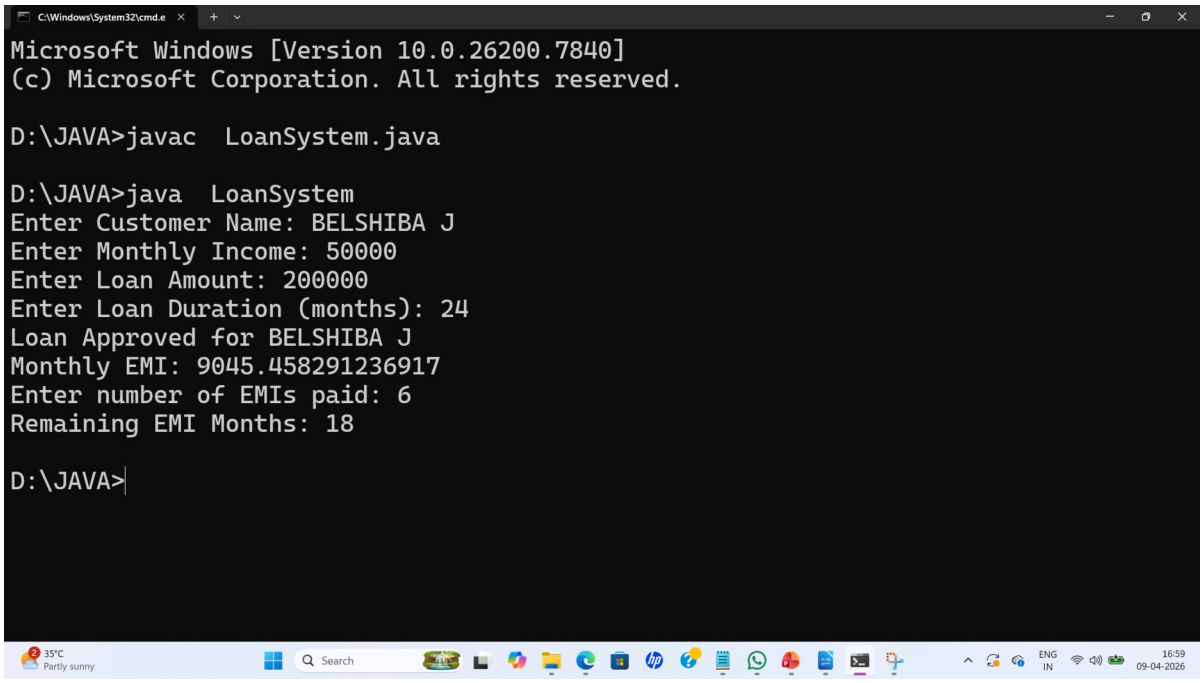
        obj.getDetails();
        obj.checkEligibility();
        obj.showEMI();
        obj.trackPayment();
    }
}
```

6. INPUT AND OUTPUT SCREENS

INPUT SCREEN



OUTPUT SCREEN



7. DISCUSSION & CONCLUSION

7.1 Discussion

The Loan Eligibility Verification and EMI Payment Tracking System is designed to automate and streamline the loan approval and repayment monitoring processes. The system integrates multiple functionalities such as:

- The system evaluates applicants based on predefined criteria like income, credit score, existing liabilities, and repayment capacity.
- It reduces human error and bias by applying consistent rules for eligibility, thereby ensuring fairness and efficiency in decision-making.
- Advanced algorithms can flag high-risk applicants, helping financial institutions mitigate potential defaults.

7.2 Conclusion

The Loan Eligibility Verification and EMI Payment Tracking System provides a robust solution to automate the loan processing lifecycle. By combining eligibility assessment with real-time EMI monitoring, the system:

- Enhances operational efficiency for financial institutions.
- Minimizes human error and potential biases in loan approval.
- Improves borrower experience through transparency and timely notifications.
- Contributes to risk mitigation by tracking repayment behavior.

8. REFERENCES

BOOKS

- *Credit Risk Management: Basic Concepts* by Tony Van Gestel and Bart Baesens — Covers **credit scoring and credit risk analysis**, which is foundational for understanding how eligibility is assessed in loan systems.
- *The Principles of Banking* by Moorad Choudhry — Provides a broad overview of banking operations, including **lending policy, loan products, and financial management principles** that support systems like loan eligibility checks.

WEBSITES

https://www.bankbazaar.com

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